

Lesson 4: Electromagnet

Current, turns, core, and fair tests

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

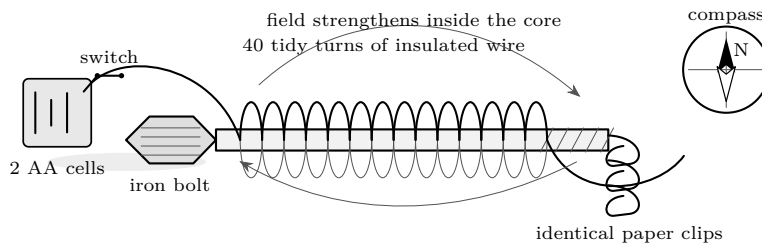
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- 2 AA cells in fused, switched holder
- 1–2 m insulated magnet wire
- Iron bolt, cardboard tube, and tape
- 12 identical paper clips in a cup
- Compass and multimeter
- Ruler, timer, and data sheet

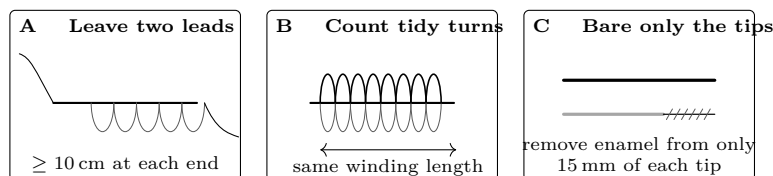
Predict first

Commit before building. Which should change lifting strength more: doubling turns or reversing battery polarity? Will replacing iron with air increase, decrease, or preserve the lift? Give a reason for each prediction.



Investigate

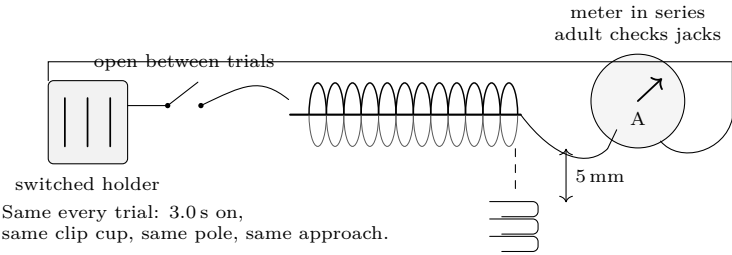
Stage 1: make a repeatable coil



1. Wind 20 close, single-layer turns around the cardboard tube. Tape the winding so neither the turn count nor winding length can change. Slide the bolt in only after checking that the tube moves freely.
2. Adult checks continuity and measures coil resistance with the battery disconnected: $R = \text{_____} \Omega$. If the meter shows open circuit, diagnose before energizing.
3. Connect the fused switched holder. Keep the master switch open except during a timed trial.

Checkpoint question. Why would comparing a loose 20-turn coil with a compact 40-turn coil change two variables at once?

Stage 2: establish the test method



- 1. Set the lower bolt pole 5 mm above a cup of identical clips. Close the switch for exactly 3.0 s, lift vertically, count the clips still held, then open the switch.
- 2. Record current during the pulse. Wait at least 30 s, touch-check for warmth, return every clip to the cup, and repeat three times.

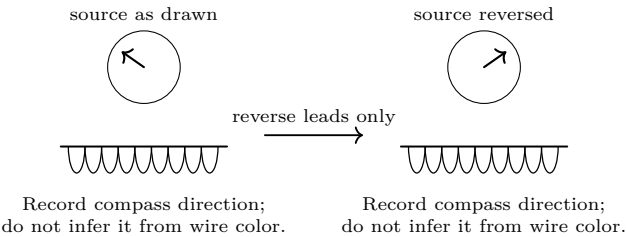
condition	turns	core	current (A)	lift 1	lift 2	lift 3
baseline	20	iron	_____	_____	_____	_____

Checkpoint questions. What varied across the three repeats? Is the spread small enough that a one-clip difference would be convincing?

Stage 3: change one variable

Run the matrix below. Rewind to 40 turns without changing winding length. For the air-core test, slide the bolt out; do not move the coil or clip cup. Reverse only the battery leads for the polarity test.

test	turns	core	polarity	mean lift	variables held fixed
A	20	iron	normal	_____	source, pulse, gap, clips, pole
B	40	iron	normal	_____	source, pulse, gap, clips, pole
C	40	air	normal	_____	source, pulse, gap, clips, coil
D	40	iron	reversed	_____	source, pulse, gap, clips, coil



Intermediate claims. Complete only after measuring:

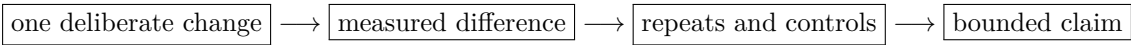
- a. Doubling turns changed mean lift from _____ to _____; circle: did / did not exceed trial-to-trial spread.
- b. Replacing iron with air changed mean lift by _____ clips while all listed controls stayed fixed.
- c. Reversing current changed compass direction from _____ to _____, while mean lift changed by _____.

If the apparatus fails

symptom	check while disconnected	likely correction
no current; no lift	enamel at both tips, switch, fuse, continuity	scrape only the lead tips; repair one open joint
large current; no lift	winding path and accidental bypass around coil	remove the short path; reconnect through every turn
results jump widely	pulse time, gap, clip type, vertical approach	reset jig; use one timer and one clip cup
coil or holder warms	pulse length, shorted turns, current	stop; disconnect; adult inspects before reuse
compass barely moves	distance, nearby steel, compass freedom	move compass closer after deenergizing; clear steel objects

Stage 4: the evidence proof

Choose the strongest supported conclusion and write a four-link proof:



Changed variable: _____ Measured difference: _____

Repeat spread: _____ Therefore the evidence supports: _____

Final challenge. What result would make you reject your conclusion even if it matched your original prediction?

Explain from the evidence

Current in every turn produces a magnetic field. With the turns wound in the same sense, their axial fields reinforce. The iron core magnetizes and offers a strong path for flux, so it can increase the force at the bolt pole. Reversing current reverses north and south; it does not, by itself, predict a large strength change. Turn count is not the only influence: current, geometry, core, gap, pulse time, and heating can all alter the observation. That is why repeats and controlled variables are part of the physical argument, not paperwork after it.

Change one thing

Graph mean lift against 20, 30, and 40 turns. Include a vertical uncertainty bar spanning the lowest to highest trial. Before calling the graph a “turns” result, inspect whether current changed as added wire increased resistance. State the narrower claim your measurements actually justify.

Evidence record	
Prediction	_____
One change	_____
Observation	_____
Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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